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A Case Study of Microplastics Detection through an Integrated STEM Framework

iSTEM Education | Environmental AI

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Introduction

Microplastics (<5mm) are plastic particles found in oceans, soil, sediments, and human tissue

Traditional detection (FTIR, Raman spectroscopy) is accurate but cost-prohibitive for undergraduate labs

Manual visual microscopy is cheap but subjective, tedious, and error-prone

This study proposes a 4-phase iSTEM pipeline using a low-cost USB microscope and 4 deep learning models

Students collect real samples, image them, train AI models, and statistically validate results

The Problem: Why Microplastics Matter

- Microplastics (<5mm) found in oceans, rivers, soils, sediments, and human tissue worldwide
- Marfella et al. (2024): particles in carotid plaques linked to 4.53x higher cardiovascular risk
- Zhang et al. (2025): disrupts mitochondrial energy production, triggers immune response
- FTIR & Raman spectroscopy: accurate but too expensive for most undergraduate labs
- Manual microscopy: cheap but subjective, tedious, and error-prone



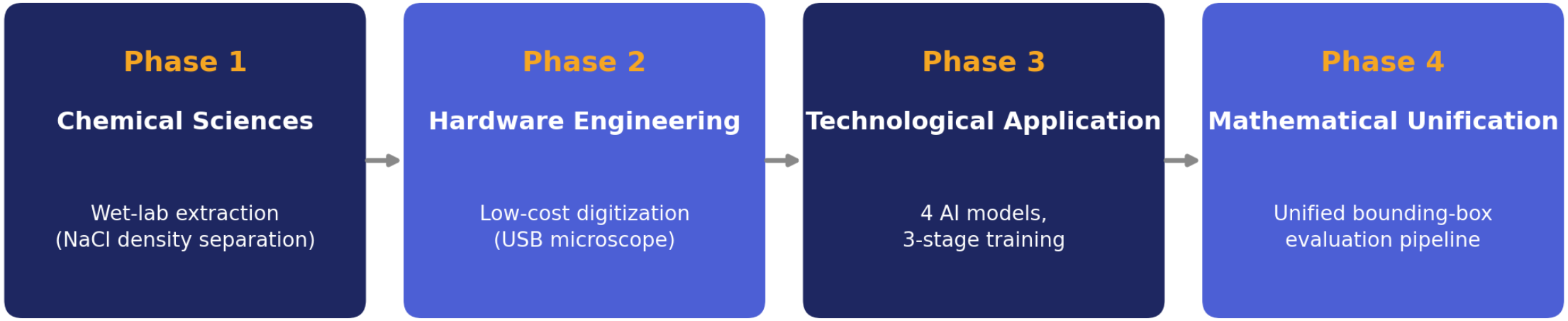
The Research Gap: What Was Missing

- No low-cost pipeline: existing AI tools require expensive equipment or clean lab-quality images
- No unified comparison: segmentation and detection models use incompatible metrics (mIoU vs mAP)
- No iSTEM integration: no framework combines wet-lab chemistry, hardware, and AI for undergraduates

Our Approach

- A 4-phase iSTEM pipeline using a low-cost USB microscope
- A unified bounding-box evaluation enabling fair comparison across all 4 AI models

Our iSTEM Framework: 4-Phase Pipeline



Phase 1 & 2: Wet-Lab Extraction & Imaging

Phase 1: Chemical Sciences (Wet-Lab)

- Sample: Bang Saen Beach sand, Chonburi, Thailand
- Saturated NaCl density separation — plastics float, sediment sinks
- Fine mesh filtration, air drying, and non-destructive optical confirmation

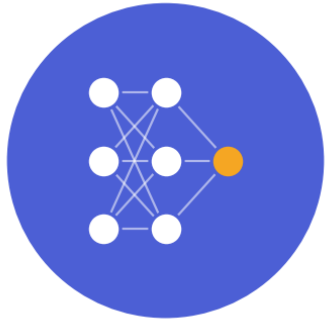


Phase 2: Hardware Engineering (Low-Cost Digitization)

- Consumer 40x USB microscope with dual LED illumination
- 4032x3024 px images, resized to 1920x1080 for processing
- Manual annotation using Roboflow (bounding boxes + masks)



Phase 3: AI Models & 3-Stage Training Strategy



- Object Detection: Faster R-CNN (two-stage) and YOLOv5 (single-stage)
- Segmentation: U-Net (semantic) and Mask R-CNN (instance)
- Sub-Phase 1 — Public Dataset: "panats-mpproject" — learn basic shapes and edges
- Sub-Phase 2 — Local Fine-Tuning: 50 Bang Saen beach microscopy images
- Sub-Phase 3 — Domain Adaptation: 30 curated images for USB microscope glare & aberration

Phase 4: Evaluation Metrics

- Precision, Recall, and F1-Score computed at an IoU threshold ≥ 0.5
- All 4 models evaluated on identical terms — a fair, apples-to-apples comparison

Contamination Percentage (Cp)

- Cp = total detected particle area ÷ total image area × 100
- Turns raw AI detections into a usable environmental severity metric

Phase 4: The Evaluation Problem

The Problem

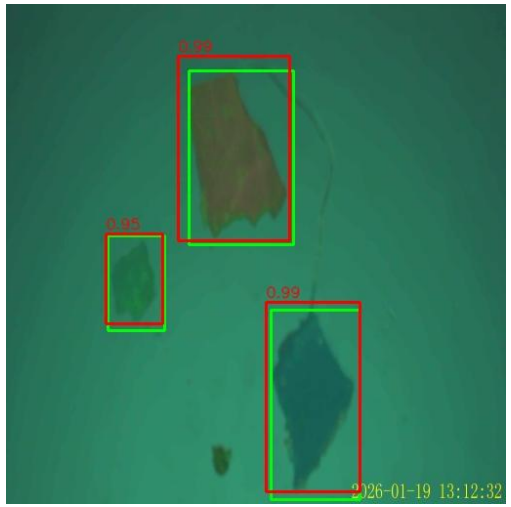
- Segmentation models output pixel masks; detection models output bounding boxes — they cannot be compared directly

Our Solution: convert every model's output into a bounding box using a 3-step process

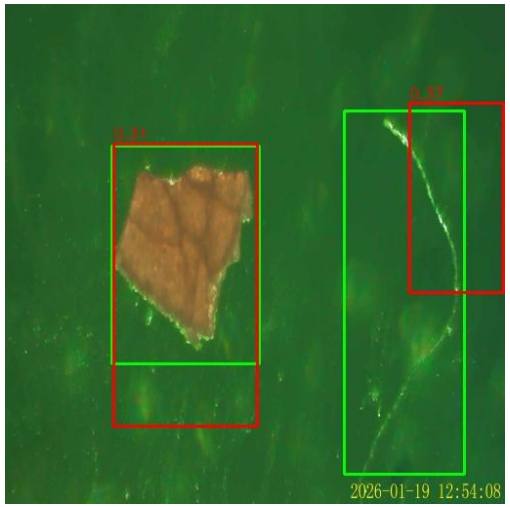
- Step 1 — Morphological Opening (erosion then dilation): removes small noise specks
- Step 2 — Morphological Closing (dilation then erosion): fills gaps in fragmented masks
- Step 3 — Contour detection: extracts the minimum enclosing bounding rectangle

Qualitative Detection Results

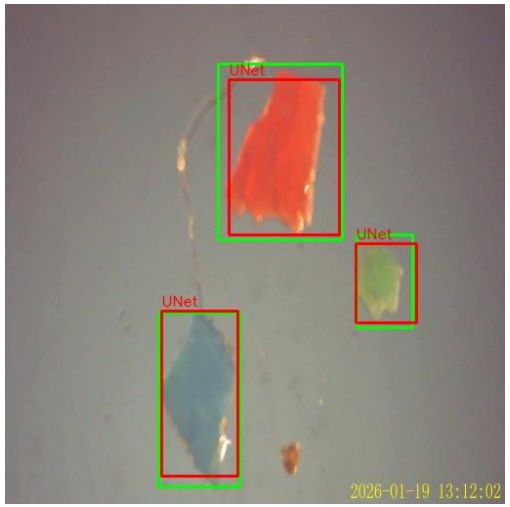
Ground truth (green) vs. predicted (red) bounding boxes on Bang Saen beach microplastic samples



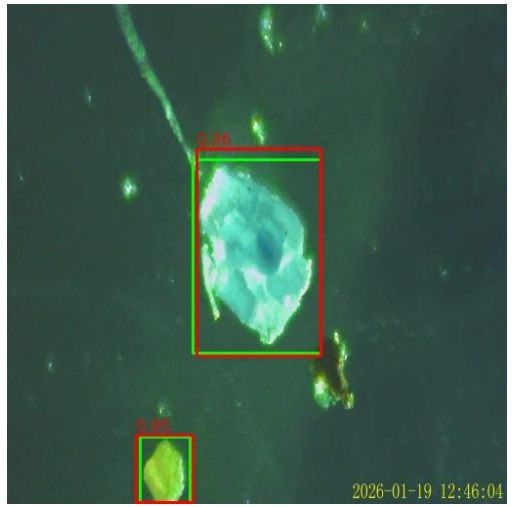
Faster R-CNN
Reliable detection across varied particle shapes and sizes



YOLOv5
High recall, with occasional false positives from glare

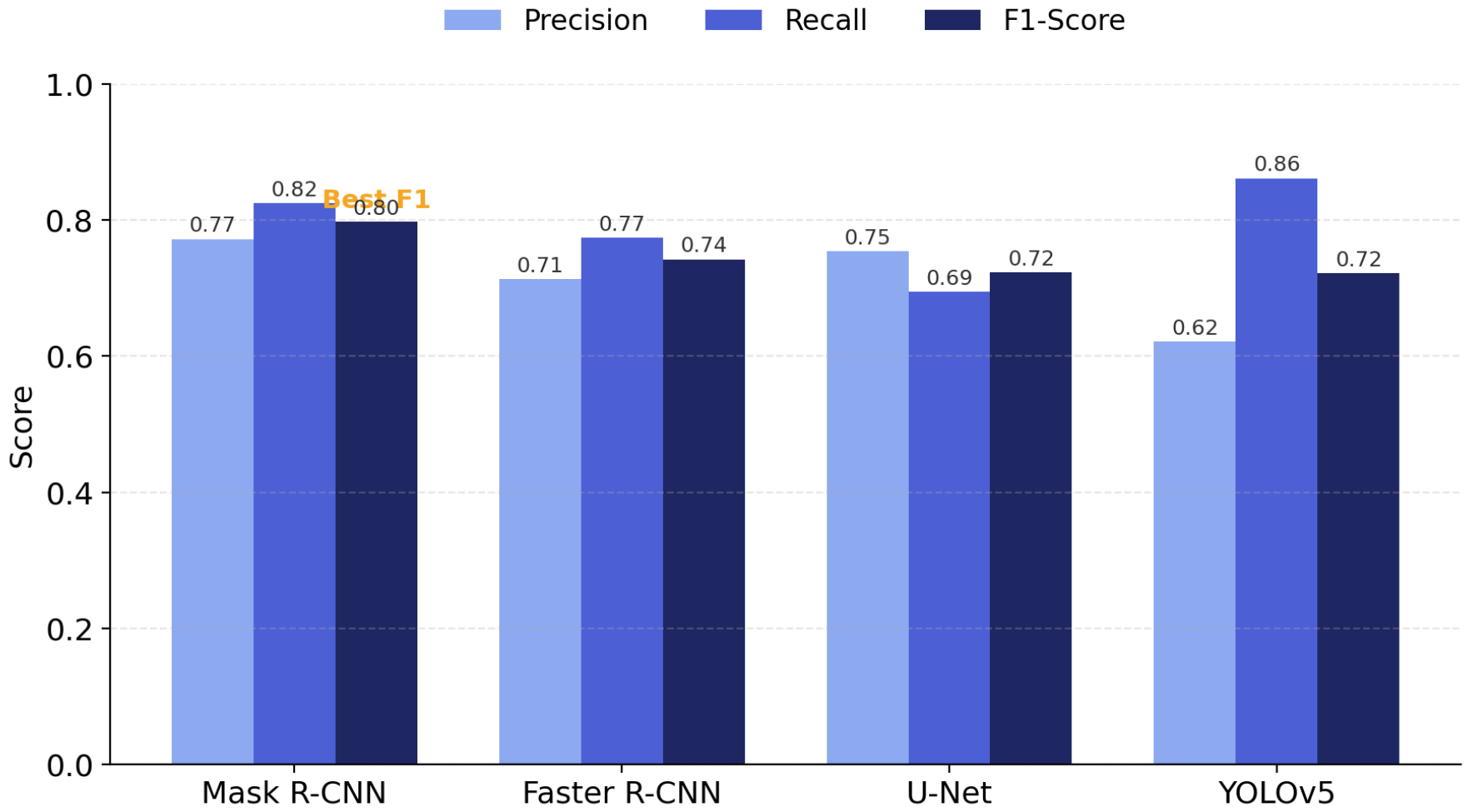


U-Net
Precise boundaries via morphological post-processing



Mask R-CNN
Best overall balance; strongest noise suppression

Results: Model Performance Comparison



What the Results Tell Us: Model Trade-offs

- Mask R-CNN (Best): F1 = 0.7978 — instance segmentation backbone suppresses optical glare noise
- Faster R-CNN (2nd): F1 = 0.7425 — two-stage design reliably catches diverse microplastic shapes
- U-Net: highest precision (0.7546) but misses smaller particles (recall 0.6949) — best used as a filter
- YOLOv5: highest recall (0.8619) but lowest precision (0.6215) — catches everything, including glare

What the Results Tell Us: The Bigger Picture

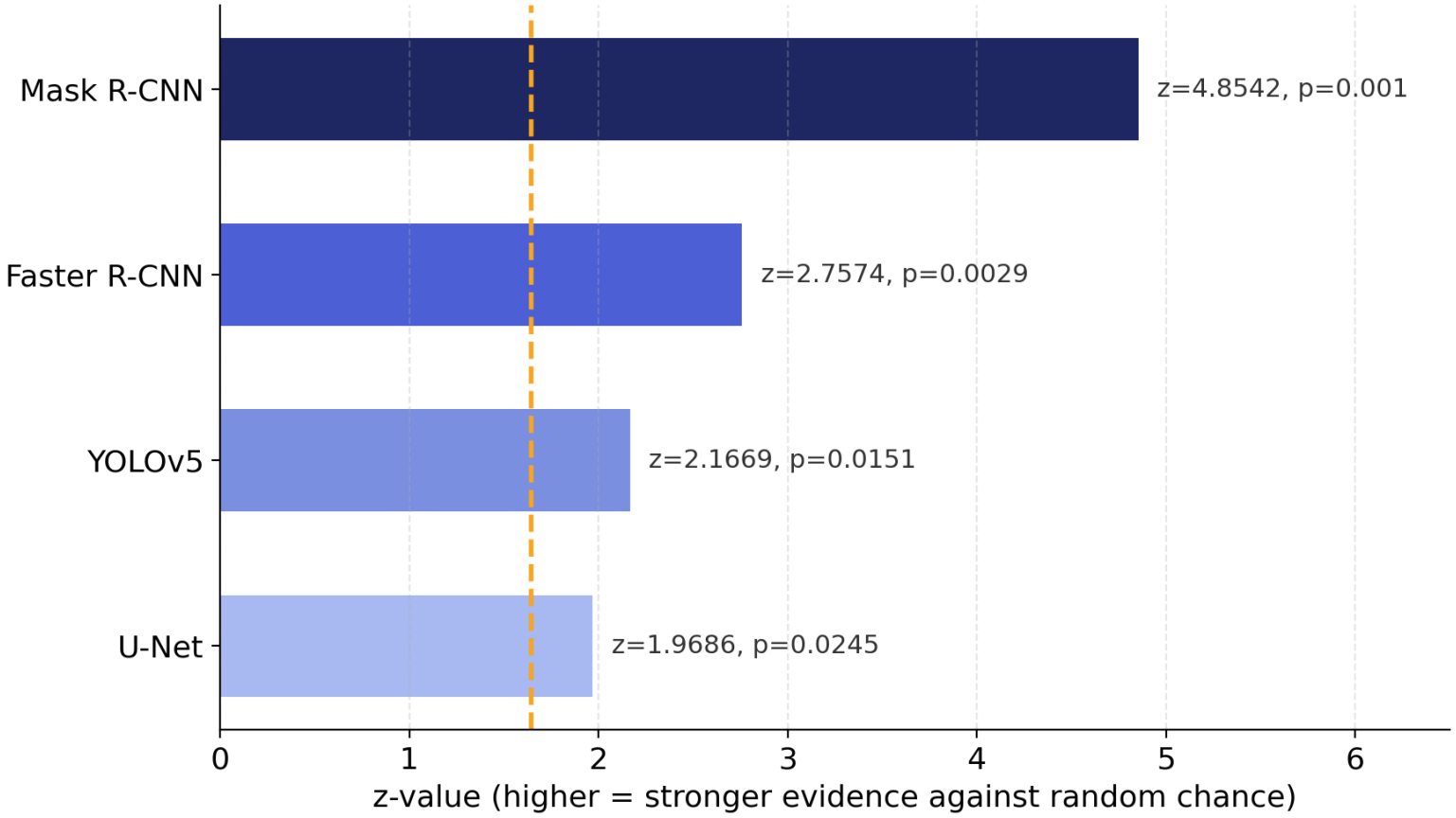
Contamination Percentage (Cp)

- U-Net and Faster R-CNN produced stable Cp medians of 8–11% across the holdout beach samples

Pedagogical Insight

- Each model teaches a real engineering trade-off: sensitivity vs. specificity vs. balance

Statistical Validation



Statistical Validation: What This Proves

- All 4 models statistically exceed the 50% random-chance baseline (right-tailed z-test, $\alpha = 0.05$)
- Negative controls: crushed glass particles — models were briefly “tricked” by optical translucency, as expected
- Deep learning is viable for microplastics detection even in resource-constrained educational settings

Educational Outcomes: What Students Learned



Reflective practice from the 3 IT student co-authors (qualitative, n=3)

- Cross-disciplinary challenge: real wet-lab chemistry, specific gravity calculations, and filtration
- Hardware reality check: USB microscope glare taught that models are only as good as their input data

Educational Outcomes: Tangibility & Limitations

- Tangibility of the problem: collecting Bang Saen beach sand made a global crisis feel personally urgent
- Computing their own Cp metric connected AI output to real environmental meaning

Limitations acknowledged

- Optical ID only (no FTIR/Raman); n=3 reflective sample creates a precedent case study

Research Contributions (1/2)

Educational Contribution

- First iSTEM framework integrating wet-lab chemistry, low-cost hardware, and AI for environmental monitoring

Technical Contribution

- Deployed and compared 4 deep learning architectures under identical low-cost conditions via 3-stage training

Research Contributions (2/2)

Methodological Contribution

- Unified bounding-box evaluation enabling cross-paradigm comparison; novel Cp metric linking AI output to environmental severity

Research Contribution

- Empirical evidence via z-testing: all 4 models exceed the 50% random-chance baseline ($p < 0.05$)

Conclusion

- Demonstrated a complete environmental AI pipeline using only low-cost, commercial off-the-shelf hardware
- Achieved best F1 = 0.7978 (Mask R-CNN); all 4 models statistically exceed random chance ($p < 0.05$)
- Established a unified bounding-box evaluation enabling fair cross-paradigm comparison in resource-constrained settings
- Provided a replicable blueprint for merging computing education with real environmental science

Limitations & Future Work

Limitations

- Optical ID only — no FTIR/Raman polymer validation
- USB microscope detects only particles larger than 8 micrometers
- Reflective sample n=3 (first three co-authors); creates a precedent case study, not yet generalizable

Future Work

- Add native mIoU/Dice metrics for segmentation models
- Compare few-shot domain adaptation vs. CLAHE augmentation
- Pre/post intervention surveys for stronger pedagogical claims

Thank You

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